

E-Commerce Analytics Dashboard Report

Project Title

E-Commerce Analytics Dashboard using Power BI, MySQL, and Python

1. Introduction

The E-Commerce Analytics Dashboard project was developed to analyze business performance, customer behavior, product performance, and customer churn using interactive dashboards. The project focuses on transforming raw e-commerce data into meaningful business insights that can help organizations make data-driven decisions.

The dashboard was designed using Microsoft Power BI with data stored and managed through MySQL. Data cleaning and preprocessing were performed using Python.

This project demonstrates practical business intelligence skills including data cleaning, data modeling, DAX calculations, dashboard designing, and business analysis.

2. Objectives of the Project

The main objectives of this project are:

- Analyze overall sales and revenue performance
- Identify top-performing products and categories
- Understand customer purchasing behavior
- Monitor customer churn and retention
- Create interactive business dashboards for decision-making
- Build a professional analytics portfolio project

- **3. Tools and Technologies Used**

Tool / Technology Purpose

Power BI	Dashboard creation and visualization
MySQL	Data storage and management
Python	Data cleaning and preprocessing

4. Dataset Overview

The dataset contains e-commerce transactional data including:

- Customer information
- Product details
- Order information
- Sales and revenue data
- Country-wise transactions
- Product categories
- Customer purchase history

The dataset was cleaned and transformed before loading into Power BI.

5. Data Cleaning and Preprocessing

The following preprocessing steps were performed:

- Removed duplicate records
- Handled missing values
- Corrected inconsistent data entries
- Converted date columns into proper datetime format
- Created calculated columns for analytics
- Imported cleaned data into MySQL database
- Connected Power BI with MySQL for dashboard creation

6. Data Modeling

A relational data model was created in Power BI to establish relationships between different tables.

Key tables used in the project:

- Orders Table
- Customers Table
- Products Table
- Order Items Table

Relationships were created using primary and foreign keys to support accurate analysis and filtering.

7. DAX Measures and KPIs

Several DAX measures were created to calculate important business KPIs.

Main KPIs

- Total Revenue
- Total Orders
- Total Customers
- Average Unit Price
- Repeat Customer Percentage
- Churn Customers
- Active Customers
- Total Products
- Total Categories

8. Dashboard Pages

The dashboard consists of four major analytical pages.

8.1 Executive Dashboard

The Executive Dashboard provides a high-level business overview.

Features

- Revenue KPI cards
- Total Orders analysis
- Monthly sales trend analysis
- Country-wise sales visualization
- Top-selling products analysis

Business Insights

- Revenue trends can be monitored over time
- Top-performing countries contribute major revenue
- Certain products generate significantly higher sales

8.2 Customer Intelligence Dashboard

The Customer Intelligence Dashboard focuses on customer behavior and segmentation.

Features

- Customer segmentation analysis
- Repeat customer percentage
- Revenue contribution by customer segment
- Customer purchase behavior analysis
- Customer distribution visualizations

Business Insights

- Repeat customers contribute significant business revenue
- Customer segmentation helps identify high-value customers

Certain customer groups generate higher purchase frequency

8.3 Churn Analysis Dashboard

The Churn Analysis Dashboard identifies customer retention and churn patterns.

Features

- Churn customer KPIs
- Active vs churned customer analysis
- Retention monitoring table
- High-risk customer identification
- Churn percentage analysis

Business Insights

- High recency customers are more likely to churn
- Customer retention trends can be monitored effectively
- Identifying high-risk customers helps improve retention strategies

8.4 Product Analytics Dashboard

The Product Analytics Dashboard analyzes product performance and category trends.

Features

- Product revenue analysis
- Category-wise sales analysis
- Product performance comparison
- Quantity vs revenue analysis
- Geographic product sales visualization

Business Insights

- Certain categories contribute maximum revenue
- High-performing products drive overall sales growth
- Product-level analysis helps improve inventory and marketing decisions

9. Dashboard Features

The dashboard includes several interactive and advanced features:

- Interactive slicers and filters
- Dynamic KPI cards
- Drill-down analysis
- Multiple dashboard pages
- Responsive visualizations
- Geographic analysis using maps
- Customer churn analysis
- Product performance tracking

10. Challenges Faced

During the project development process, several challenges were encountered:

- Handling null and inconsistent data
- Creating proper table relationships

- Managing DAX calculations
- Designing professional dashboard layouts
- Improving visual clarity and dashboard performance

These challenges were solved through data preprocessing, optimization, and dashboard redesign.

11. Skills Demonstrated

This project demonstrates the following technical and analytical skills:

Technical Skills

- Power BI Dashboard Development
- Data Modeling
- DAX Calculations
- SQL Database Management
- Python Data Cleaning
- Data Visualization

Analytical Skills

- Business Intelligence
- Customer Analysis
- Churn Analysis
- Product Performance Analysis
- KPI Development
- Data-Driven Decision Making

12. Conclusion

The E-Commerce Analytics Dashboard successfully transforms raw business data into meaningful and interactive business insights. The project demonstrates practical knowledge of business intelligence tools and analytics techniques.

By combining Power BI, MySQL, Python, and DAX, the dashboard provides valuable analysis related to sales performance, customer behavior, product analytics, and customer churn.

This project also serves as a strong portfolio project demonstrating end-to-end data analytics and dashboard development skills.

13. Future Improvements

Possible future enhancements for the project include:

- Real-time data integration
- Sales forecasting using machine learning
- Advanced customer segmentation
- Inventory optimization analysis
- Mobile dashboard optimization
- Automated report generation

14. Final Outcome

The final output of the project is a professional multi-page interactive Power BI dashboard capable of providing meaningful business insights for e-commerce decision-making.

The project demonstrates strong understanding of:

- Data analytics workflow
- Business intelligence concepts
- Dashboard development
- Interactive reporting
- Analytical thinking